MILITARY SPECIFICATION

CONTROL, DEPTH, TORPEDO C-7821/A

This specification has been approved by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. The equipment covered by this specification shall be for airborne use and shall provide for inflight manually selecting depth settings of the Mk 44 Mod0 or Mod 1 Torpedo or the search depth and ceiling settings of the Mk 46 Mod 0, Mod 1 or Mod 2 Torpedo prior to their release from the carrying aircraft. In addition, the equipment shall be capable of selecting the SNAKE mode gyro settings of the Mk 46 Mod 1 and the HATS mode gyro settings for the Mk 46 Mod 2 Torpedos. Also, a visual indication that the torpedo setting matches the selected setting shall be provided.

1.2 Classification. This equipment covered by this specification shall consist of the following items:

Type Designation	Applicable Paragraph
Control, Depth, Torpedo C-7821/A	3.5.1

1.3 Associated Equipment. This equipment shall operate with the associated equipment listed in 6.7.

2. APPLICABLE DOCUMENTS

2.1 General. The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-W-5088	Wiring Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment

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MIL-P-24014	Preclusion of Hazards from Electromagnetic Radiation to Ordnance, General Requirements for
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Panels, Information, Integrally Illuminated
MIL-E-1755S	Electronic and Electrical Equipment, Accessories and Repair Parts; Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

Naval Air Systems Command

STANDARDS

Military

MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of
AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for
AR-34	Failure Classification for Reliability Testing, General Requirements for

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PUBLICATIONS

Naval Air Systems Command

EI-604	Avionics Installation Instructions for Torpedo Depth Control C-7821/A
ET-604	Preflight and Flight Test Instructions for Torpedo Depth Control C-7821/A

2.1.1 Availability of Documents.

a. When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Preproduction. This specification makes provision for preproduction testing.

3.2 Parts and Materials. In the selection of parts and materials fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Microelectronic Technology shall be considered and microelectronic items shall conform to requirements specified herein.

b. Other parts and materials requirements shall conform to MIL-E-5400.

c. Nonrepairable subassemblies shall be used in accordance with AR-10 and as outlined in MIL-E-5400.

d. When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval. Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diodes) other than microelectronic devices shall be obtained as outlined in MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.2.2 Microelectronic Modular Assemblies. When used, Microelectronic Modular assemblies shall meet the requirements of AR-5. Conformal coatings, encapsulants, embedments or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.2.3 Modules. The electronic portions of the equipment shall be functionally modularized in accordance with AR-10.

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3.3 Design and Constuction. The equipment shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight. The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 4 pounds.

3.3.2 Reliability. The contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated. in the contract or order.

3.3.2.1 Operational Stability. The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 3500 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life. The equipment shall have a total operating life of 6000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF). The equipment, including any built-in-test provisions, shall have 3500 hours of mean (operating] time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.3 Cabling and Connections.

3.3.3.1 Cables and Connectors. The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.3.3.2. Interconnection Cabling. The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Control Panels. All rack or console mounted control panels shall conform to the applicable requirements of MIL-C-6781. The configuration of all panels must be approved by the procuring activity prior to preproduction testing.

3.3.5 Interchangeability. The equipment shall meet the interchangeability requirements of MIL-E-5400.

3.3.6. Interference Control. The generation Of radio interference by the equipment and the vulnerability of the equipment to radio interference shall meet the requirements of MIL-STD-461.

3.3.7 Provisions for Maintainability. The Maintainability Program, built-in-test features, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in AR-10.

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3.3.8 Nomenclature, Nameplates, and Identification Marking. Nomenclature and serial number assignment, nameplate approval, and identification marking shall be in accordance with MIL-N-18307.

3.3.9 Standard Conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory-bench tests.

3.3.10 Service Conditions. The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 1 equipment, except as modified herein.

3.3.10.1 Vibration. The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve IV of MIL-T-5422 except that the amplitude shall be limited to 5g's.

3.3.11 Warm-Up Time. No warm-up shall be required.

3.3.12 Input Electrical Power.

3.3.12.1 Operating Power. The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall not exceed the specified amounts.

a. AC Power (Single Phase), 115 V, Category B, 10 VA

b. DC Power, 28 V, Category B;

(1) 3 watts when on but not making depth selection.

(2) 68 watts typical when making a depth selection and for a period of 0.25 second maximum after making the selection.

(3) 139 watts maximum when making a depth selection and for a period of 0.25 second maximum after making the selection.

3.3.12.2 Lighting Power. Input power for lighting shall require not more than 0.400 amp at 28 volts.

3.3.12.3 Degraded Performance. Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation, Operation shall return to normal with no resulting damage to the equipment.

3.4 Performance. The performance of the equipment shall be as set forth below. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variations are specified.

3.5 Detail Requirements.

3.5.1 Torpedo Control C-7821/A The Torpedo Depth Control shall meet the requirements stated in the following subparagraphs.

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3.5.1.1 Controls. The equipment shall contain the following controls which shall be access the operator:

3.5.1.1.1 Mk 46 Ceiling/Mk 44 Floor control. A 12 Position, 30° detent rotary switch shall be used for selecting the Mk 46 CEIL, Mk 46 SNAKE Mk 46 HATS, or Mk 44 FL. The settings shall appear as panel markings and be located as follows:

Setting	Location from 12:00 O'Clock Position
Mk 46 HATS	Counterclockwise 75°
Mk 46 SNAKE	Counterclockwise 45°
Mk 46 P20	Counterclockwise 150°
Mk 46 P50	Clockwise 15°
Mk 46 A20	Clockwise 45°
Mk 46 A50	Clockwise 75°
Mk 44 OFF	Counterclockwise 105°
Mk 44 150	Counterclockwise 135°
Mk 44 250	Counterclockwise 165°
Mk 44 450	Clockwise 165°
Mk 44 650	Clockwise 135°
Mk 44 900	Clockwise 105°

3.5.1.1.2 Mk 46 Search Depth/MK 44 Initial Search Depth. A 12 position, 30° detent, rotary switch shall be used for selecting the Mk 46 SRCH DEPTH or Mk 44 ISD settings. The setting shall appear as panel markings and be located as follows:

Setting	Location from 12:00 O'Clock Position
125	Counterclockwise 75°
275D	Counterclockwise 45°
275	Counterclockwise 15°
500	Clockwise 15°
750	Clockwise 45°
1000	Clockwise 75°
50	Counterclockwise 105°
150	Counterclockwise 135°
250	Counterclockwise 165°
450	Clockwise 165°
65	Clockwise 135°
900	Clockwise 105°

3.5.1.1.3 Mk 46 Gyro. A 12 position, 30° detent, rotary switch shall be used for selecting the Mk 46 gyro settings. The setting from PORT 95° to STBD 95° shall be marked Mod 2 ONLY and shall require that the switch must be pulled to enter these positions. The settings shall appear as panel markings and be located as follows:

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Setting	Location from 12:00 O'Clock Position	
0	At 12:00 O'Clock	
PORT 35°	Counterclockwise	30°
PORT 70°	Counterclockwise	60°
PORT 95°	Counterclockwise	90°
PORT 120°	Counterclockwise	120°
PORT 150°	Counterclockwise	150°
180°	Counterclockwise	180°
STBD 150°	Clockwise	150°
STBY 120°	Clockwise	120″
STBD 95°	Clockwise	90°
STBD 70°	Clockwise	60°
STBD 35°	Clockwise	30°

3.5.1.2 Indications. The equipment shall contain the following indicator visible to operator.

3.5.1.2.1 Ready. A green indicating lamp marked RDY.

3.5.1.2.2 Fail. A amber indictating lamp marked FAIL.

3.5.1.3 Power Control. The "OFF" positions on the Mk 46 CEIL and Mk 44 FL switch shall control all power to all circuitry (except the illuminated front panel lamp circuit) in the torpedo depth control, and to all the related circuitry in the associated equipment (torpedo). In addition, if settings are attempted for Mk 44 and MK 46 depths simultaneously, power will not be applied to the torpedo depth control (except for front panel lamps) or associated equipment.

3.5.1.4 Mk 46 Ceiling/MK 44 Floor Mode. The equipment shall provide a null circuit and one [control or reference) branch of a self balancing bridge network of which the second (controlled) branch is part of the accessory equipment. The voltage. provided to the bridge by the control branch shall be manually selected; and shall provide a voltage reference to the null circuit for comparison with the Mk 46 ceiling or Mk 44 floor voltages provided by the accessory equipment. The voltage provided to the bridge by the controlled branch (of the accessory equipment) will be selected by a rotary stepping switch. The equipment shall provide a drive voltage to the stepping switch of the accessory equipment whenever the Mk 46 ceiling or Mk 44 floor voltages from the accessory equipment does not match (or equal) the reference voltage selected.

3.5.1.4.1 Mk 46 Ceiling/Mk 44 Floor Input Signals. The equipment shall accept at pin J1-20 any voltage at or within the following levels:

MK 46 Ceiling	MK 44 Floor	Voltage
		E
P20	900	0.784E to 0.812E
P50	650	0.576E to 0.624E
A20	450	0.376E to 0.424E
A50	250	0.185E to 0.216E
	150	

The voltage is a function of the input voltage E supplied to pin J1-10.

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3.5.1.4.2 Mk 46 Ceiling/Mk 44 Floor Output Signal. For any combination of the equipment Mk 46 CEIL/Mk 44 FL switch position voltage and the reference voltage, provided by the accessory equipment, except matching combinations, the equipment shall be capable of providing at pin J1-12 up to 2.5 amperes to an inductive load at a voltage 1.0 volt less than the input power supplied to pin J1-10. For all matched combinations within 1.8 volts of each other, the output voltage at pin J1-12 shall be zero.

3.5.1.5 Mk 46 Search Depth/Mk 44 Initial Search Depth Mode. The equipment shall provide a null circuit and one (control or reference) branch of a self balancing bridge network of which the second (controlled) branch is part of the accessory equipment. The voltage provided to the bridge by the control branch shall be manually selected, and shall provide a voltage reference to the null circuit for comparison with the Mk 46 SRCH DEPTH or Mk 44 ISD voltages provided by the accessory equipment. The voltage provided to the bridge by the controlled branch (of the accessory equipment) will be selected by a rotary stepping switch. The equipment shall provide a drive voltage to the stepping switch (of the accessory equipment) whenever the Mk 46 SRCH DEPTH or Mk 44 ISD voltages from the accessory equipment does not match (or equal) the reference voltage selected.

3.5.1.5.1 Mk 46 Search Depth/Mk 44 Initial Search Depth Input Signals. The equipment shall accept at pin J1-8 any voltage at or within the following levels:

Mk 46 SRCH DEPTH	Mk 44 IDS	Voltage
125	50	0.131E to 0.156E
275D	150	0.266E to 0.307E
275	250	0.404E to 0.453E
500	450	0.547E to 0.596E
750	650	0.693E to 0.734E
1000	900	0.844E to 0.869E

The voltage is a function of the input voltage E supplied to pin J1-10.

3.5.1.5.2 Mk 46 Search Depth/Mk 44 Initial Search Depth Output Signal. For any combination of the equipment Mk 46 SRCH DEPTH/MK IDS switch positions, the equipment shall be capable of providing at pin J1-27 up to 2.5 amperes to an inductive load at a voltage 1.0 volt less than the input power supplied to pin J1-10. For all matched combinations within 1.8 volts of each other, the output voltage at pin J1-21 shall be zero.

3.5.1.6 SNAKE Mode. When the Mk 46 CEIL/Mk 44 FL switch is placed in the SNAKE position, the Mk 46 GYRO switch shall be energized only for PORT 35, PORT 70, 0, STBD 35 and STBD 70 positions. The operation of the Mk 46 GYRO swich shall have no effect in any other positions.

3.5.1.6.1 The equipment shall provide a null circuit, and one (control or reference) branch of a self balancing bridge network of which the second (controlled} branch is part of the accessory equipment. The voltage provided to the bridge by the control branch shall be manually selected and shall provide a voltage reference to the null circuit for comparison with the SNAKE mode voltages provided by the accessory equipment. The voltage provided to the bridge, by the controlled branch (of the accessory equipment), will be selected by a rotary stepping switch. The equipment shall provide a drive voltage to the stepping switch (of the accessory equipment} whenever the SNAKE Mode voltage from the accessory equipment does not match (or equal) the reference voltage selected.

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3.5.1.6.2 SNAKE Mode Input Signals. The equipment shall accept at pin J1-20 any voltage at or within the following levels:

Mk 46 GYRO	Voltage
0	0.576E to 0.624E
STBD 35	0.784E to 0.812E
STBD 70	E
PORT 70	0.185E to 0.216E
PORT 35	0.376E to 0.424E

The voltages are a function of voltage E supplied to pin J1-10.

3.5.1.6.3 SNAKE Mode Output Signal. For any combination of eqiupment Mk 46 GYRO switch positions, the equipment shall be capable of provididng at pin J1-27 up to 2.5 amperes to an inductive load at a voltage 1.0 volt less than the input power supplied to pin J1-10. The output voltage at pinJ1-27 shall be zero for all matches combinations within 1.8 volts of each other.

3.5.1.7 HATS Mode. When the Mk 46 CEIL/Mk 44 FL switch is placed in the HATS position, the gyro switch shall be energized.

3.5.1.7.1 The equipment shall provide a null circuit, and one (control or reference) branch of a self balancing bridge network of which the second (controlled) branch is part of the accessory equipment. The voltage provided to the bridge by the control branch shall be manually selected and shall provide a voltage reference to the null circuit for comparison with the HATS mode voltages provided by the accessory equipment. The voltage provided to the bridge, by the controlled branch (of the accessory equipment), will be selected by a rotary stepping switch. The equipment shall provide a drive voltage to the stepping switch (of the accessory equipment) whenever the HATS Mode voltage from the accessory equipment does not match (or equal) the reference voltage selected.

3.5.1.7.2 HATS Mode Input Signals. The equipment shall accept at pin J1-20 any voltage at or within the following levels:

Mk 46 GYRO	Voltage
0	0.576E to 0.624E
STBD 35	0.784E to 0.812E
STBD 70	E
STBD 95	0.266H to 0.307H
STBD 120	0.131H to 0.156H
STBD 150	O.404H to 0.453H
180	0.547Hto 0.596H
PORT 150	O.693H to 0.734H
PORT 120	H
PORT 95	0.844Hto 0.869H
PORT 70	0.185E to 0.216E
PORT 35	0.376E to 0.424E

The voltages are a function of voltage E supplied to pin J1-10 and voltage H (-24 ± 1.2 volts dc) supplied by the equipment.

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3.5.1.7.3 HATS Mode Output Signal. The equipment shall be capable of providing at pin J1-27 up to 2.5 amperes to an inductive load at a voltage 1.0 volt less than the input power supplied to pin J1-10 for any combination of equipment Mk 46 GYRO switch positions. The output voltage at pin J1-27 shall be zero for all matches combinations within 1.8 volts of each other.

3.5.1.8 SNAKE and HATS Logic. The equipment shall provide the following voltages to the J1 pins indicated for the nodes shown.

MODE	J1-5	J1-6	J1-24
SNAKE	0	+24V	+24V
HATS	-24V	+24V	+24V
All other switch positions.	+24V	0	0

3.5.1.9 Current Limiting. The -24 voltage supplied by the equipment to J1-5 shall be limited to a current of 400 ma maximum.

3.5.1.10 READY Indication. The READY indicator shall be a green incandescent lamp with press-to-test and mechanical dimming capabilities The lamp shall be capable of operating on the voltage provided by pin J1-10. The lamp shall light when all the following conditions are met:

a. Both bridge circuits in the equipment are balanced and the voltages at pins J1-12 and J1-27 are zero.

b. The resistance at pin J1-18 is zero ohms.

3.5.1.11- FAIL Indication. The FAIL indication shall be an amber incandescent lamp with press-to-test and mechanical dimming capabilities The lamp shall be capable of operating on the voltage provided by pin J1-10 The lamp shall light if any of the following conditions are met:

a. Either bridge circuit in the equipment is unbalanced, and the voltage at pin J1-12 or J1-27 is 28 volts.

b. The resistance at pin J1-18 is 1000 ohms or greater.

It shall be impossible that the READY and FAIL lights. can be lit simultaneously.

3.5.1.11.1 Pail Interlock. When the FAIL lamp is lit a fail interlock signal shall be provided at J1-22. The signal shall be capable of providing up to 100 milliamperes to an inductive load at a voltage 1.0 volt less than the input power supplied to pin J1-10.

3.5.1.12 Form Factor. The maximum dimensions of the equipment shall not exceed 5.750 inches wide, by 4.500 inches high, by 5.120 inches deep (not including airframe connector and cabling).

3.5.1.13 Electrical Connections. Connections to external circuits shall be provided by receptacle type MS3140E27-50P as follows:

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Reference Designation	Pin Function
J1-5	Torpedo pin A
J1-6	Torpedo pin M
J1-8	Search depth sense
J1-9	Aircraft 400 Hz common
J1-10	Aircraft +28 VDC
J1-11	Aircraft ground
J1-12	Search depth drive
J1-14	Fire control ground
J1-18	Torpedo continuity check
J1-20	Ceiling mode sense
J1-21	Aircraft 115 VAC 400 Hz
J1-22	Fail interlock
J1-24	Gyro cage interlock
J1-26	Aircraft panel lighting
J1-27	Ceiling mode drive

3.5.1.14 Weight. The weight of the torpedo depth control shall not exceed 4 pounds.

3.5.1.15 Content. The equipment contains the following (modules):

a. Cover Assembly Top

b. Cover Assembly Bottom

c. Plate Assembly Back

3.5.2 Control, Depth, Torpedo C-7821/A. The control shall meet the applicable requirements of MIL-C-6781 for Type I control panel. The lighting plastic panel shall conform to MIL-P-7788, Class I-R, Type V with a lamp voltage of 28 volts.

4. QUALITYASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.101 Classification of Tests. Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

a. Preproduction (First Article) Tests

b. Initial Production Tests

c. Acceptance Tests

d. Life Test

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4.2 Preproduction (First Article) Tests. Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government Inspector and the procuring activity shall be advised when tests are to be conducted so that a govertient representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data. The contractor shall submit all data collected in conducting "these tests to the procuring activity for review.

4.2.2 Scope of Tests. Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of MIL-T-5422 and interference tests in accordance with MIL-STD-461.

4.2.3 Preproduction Approval. Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all-tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preparation sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments. Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection of an equipment and of equipment already accepted by the Government. It shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests. One of the first 10 production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on the equipment prim to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests. The equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material. In addition to the complete equipment submitted for initial production tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval. Approval of the initial production sample shall be by the procuring activity upon satisfactory completion

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of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the test shall be cause for deliveries of equipment index the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample. On completion of the initial production tests the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reportS showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory as applicable.Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual Tests

b. Sampling Tests

c. Reliability Assurance Tests

d. Special Tests

4.4.1 Individual Tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of Product

b. Operational Test

c. Manufacturing Run-In Test

4.4.1.1 Examination of Product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run-In Test. Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only

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those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least 6 hours of operation.

Temperature	Room ambient
Humidity	Room ambient
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6 hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the 6 hour period of operation following the 10 minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10 minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6 hour period specified above may be composed of two 3 hour periods to conform with standard working hours. This test is not required if a Reliability Assurance Test is selected which includes a test on each equipment which consumes at least 6 hours of operation.

4.4.2 Sampling Tests. Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to deselected for Sampling Test
First 10	0 (See Note)
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

Sampling Tests are not required when Reliability Assurance Tests are conducted.

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Note

If by contract action the initial production tests (4.3) are deleted, then a sample test shall be conducted on 1 equipment from the first 10 produced.

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational tests at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual, and special tests.

c. manufacturing run-in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the nuniber of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipments.

4.4.3 Reliability Assurance Tests. Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and AR-34.

4.4.3.1 Qualification Phase. Prior to the acceptance of equipments under the contract or orders a minimum of three equipments shall be tested as outlined in MIL-STD-781, under the section entitled “Qualification Phase of Production Reliability Test". The maximum number of equipments to be used shall be those listed in table 5 of MIL-STD-781. For the Qualification Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan IV shall be used.

4.4.3.2 Reliability Production Acceptance (Sampling) Phase Tests. The equipment, throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled Production Acceptance (Sampling) Phase of Production Reliability Tests”. Test level F shall be used.

4.4.3.3 Test Details. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, and preventive maintenance to be allowed during the test, shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the qualification test phase of the reliability assurance tests.

4.4.4 Special Tests. Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule. Selection of equipments for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

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4.4.4.2 Scope of Tests. Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4,5 Equipment Failure. Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure-is an isolated case or a design defect.

c. Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

d. Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test. The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual tests. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government Inspector in accordance with the following. (Equipments which have successfully passed the initial production test, sampling tests, reliability tests, or special tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions. The life test shall be conducted under the following simulated. service conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 500 ft.)
Humidity	Room ambient
AC Voltage	115 ± 5 volts (at lowest applicable frequency)
DC Voltage	27.5 ± 2.0 volts

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4.5.2 Test Periods. The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check. At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data. The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

a. Part type number

b. The circuit reference symbol number

c. The part function

d. Name of the manufacturer

e. Nature of the failure

f. The number of hours which the part operated prior to failure

4.5.4.1 Failure Report. In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the procuring activity upon. completion of the test. In this report, the conractor shall provide suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures. The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review aid approval. The right is reserved by the procuring activity or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment. Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by ‘replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing. No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

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4.9 Rejection and Retest. Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the originals shall be furnished to the Government Inspector.

5. PREPARATION FOR DELIVERY

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or-order in accordance with MIL-E-17555. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use. The Torpedo Depth Control C-7821/A is intended for use in rotary-winged aircraft to manually select, while in flight, the CIRCLE mode depth settings for Torpedo Mk 44 Mod 0, and Mod 1; Torpedo Mk 46 Mod 1, and Mod 2, and to select the SNAKE mode for Torpedo Mk 46 Mod 2. EI-604 contains installation instructions, and ET-604 contains testing instructions for the Torpedo Depth Control C-7821/A.

6.2 Performance Objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and processes other than those specified, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.3 Precedence of Documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

a. Contract. The contract shall have precedence over any specification.

b. This Specification. This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

c. Referenced Specifications. Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Ordering Data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of packaging and packing (see 5.1).

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6.5 Type Designations. The parentheses ( ) when used in the equipment type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used on nameplates, shipping records, and instruction books, as applicable.

6.6 Revisions. In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective as the asterisk ntoations and relationship to the last previous issue.

6.7 Associated Equipment. The equipment shall operate with the following associated equipment:

a. Torpedo Mk 44 Mod O

b. Torpedo Mk 44 Mod 1

c. Torpedo Mk 46 Mod O

d. Torpedo MK 46 Mod 1

e. Torpedo Mk 46 Mod 2

6.8 This specification is under the cognizance of AIR-53371D.

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-81861(AS) CONTROL, DEPTH, TORPEDO C-7821/A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

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